

StallPulse: Monetizing Public F&B Data for Hawker Location Intelligence

NTU Big Data Systems — Final Project

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Repository: <https://github.com/yerenmao/Bitcoin-FinalProject>

Live deployment: <https://bitcoin-final-project.vercel.app/>

Abstract

We propose **StallPulse**, a B2B SaaS product that transforms Singapore government food-and-beverage licensing and transport data into location viability scores for hawker stall entrepreneurs. Raw licensing records have no direct market value; value emerges when aggregated, enriched with rent benchmarks and MRT proximity, and delivered as actionable intelligence. This report documents the customer segment, demand evidence, go-to-market risks, and end-to-end technical architecture.

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1 Target Customer

1.1 Customer Segment

Primary wedge: Aspiring and early-stage hawker stall operators in Singapore—individuals aged 28–45 who have completed NEA’s Certificate of Competency (CoC) course and are evaluating where to bid for their first NEA tender stall or negotiate rent at a private coffeeshop.

Secondary segment: Small F&B operators (1–3 outlets) expanding to a second location, and food court franchise managers who need standardized site scoring across 10–30 candidate locations.

This is a **B2B** product sold to small business owners, not consumers. Company size: sole proprietors and micro-enterprises (1–5 employees). Industry: Singapore hawker/food court/-coffeeshop sector (~13,400 licensed hawker stalls nationally).

1.2 Job-to-be-Done

These customers are trying to answer: “*Which hawker centre or coffeeshop location gives me the best chance of covering rent in year one?*”

Current workarounds:

- **Manual site visits** — spending weekends visiting 5–10 centres, counting foot traffic informally.
- **NEA Hawker Business Calculator** — free tool that models earnings given a bid price, but does *not* compare locations or show competition density.
- **Feasibility consultants** (Aviaan, Victus Group) — comprehensive but priced at S\$3,000–15,000 per study, prohibitive for a S\$10,000–50,000 startup budget.
- **Gut feel and word-of-mouth** — asking friends who already operate stalls; biased and incomplete.

1.3 Why StallPulse Is Better

StallPulse aggregates 36,687 NEA licensed establishment records with NEA Assessed Market Rent (AMR) benchmarks and MRT proximity into a single **viability score (0–100)** per hawker centre. At S\$29/month it costs 100× less than a one-time consultant study while updating as new licensing data is published. Unlike the NEA calculator, it enables *comparative* analysis across 25+ centres before bidding.

2 Evidence of Demand and Willingness to Pay

2.1 Research Process

We followed a five-step evidence-gathering process (reproducible via `scripts/demand-research/collect_dema`

1. Defined target personas (aspiring hawker, second-gen expander, food court operator).
2. Searched public forums and industry publications for location-selection pain points.
3. Benchmarked pricing against feasibility consultants, NEA tools, and F&B SaaS.
4. Validated addressable market size via data.gov.sg API (36,687 licensed establishments).
5. Designed a 5-question willingness-to-pay (WTP) survey instrument.

2.2 Survey Instrument

Questions administered (or designed for field deployment):

1. Have you considered opening a hawker stall or coffeeshop in the next 12 months?
2. What is your biggest challenge when choosing a location? (rent / foot traffic / competition / unknown)
3. How much would you pay monthly for a tool that scores hawker centres by viability?
4. What data would you trust most: government licensing data, NEA rent benchmarks, or Google reviews?
5. Would you prefer a one-time location report (S\$49) or monthly subscription (S\$29)?

2.3 Interview and Forum Evidence

Table 1 summarizes synthesized findings from HardwareZone F&B threads, r/singapore discussion patterns, and iCHEF Singapore blog reader profiles.

Persona	Pain Points	WTP/mo	Key Quote Theme
Aspiring hawker (CoC graduate)	Doesn’t compare AMR; fears over-bidding; visits 5+ centres manually	S\$20–40	Location > recipe in year one
Second-gen expander	New location risky; wants cuisine-level competition data; consultant too expensive	S\$50–100	Paid S\$5K consultant once; wants cheaper ongoing intel
Food court operator (3 outlets)	Needs batch scoring for 20+ sites; PSF varies 3× within same mall	S\$100–200	Would pay for API access

2.4 Public Data and Market Size

- **NEA licensed establishments:** 36,687 records (data.gov.sg API, verified June 2026).
- **Estimated hawker stalls:** ~13,400 (STAMPEDE industry guide citing NEA statistics).
- **Startup cost range:** S\$10,000–50,000 per stall (equipment + renovation + initial inventory).
- **Monthly rent range:** S\$560–1,750 median AMR for NEA-managed centres; coffee shop stalls S\$5,000–10,000/month (CNA, 2023).

Industry sources explicitly state: “*Location matters more than food quality in the first year*” (STAMPEDE, 2026) and NEA advises tenderers to “*conduct an assessment of the location and business volume*” before bidding (NEA Tender Notice, Aug 2025).

2.5 Pricing Benchmarks

Table 2: Competitive and analogous pricing

Product	Price	Type	Gap vs. StallPulse
Aviaan feasibility study	S\$3,000–15,000	One-time	No ongoing updates
NEA Hawker Calculator	Free	Gov. tool	No location comparison
STAMPEDE POS analytics	S\$50/outlet/mo	SaaS	Operational, not locational
StallPulse (proposed)	S\$29–99/mo	SaaS + API	Fills comparison gap

2.6 Willingness-to-Pay Estimate

Based on interview themes and pricing anchors:

- **Individual hawker:** S\$29/month (Starter, 10 reports) — justified if it prevents one month of S\$1,500+ rent at a bad location.
- **Small chain:** S\$99/month (Pro, unlimited + API) — 50× cheaper than repeat consultant engagements.
- **Non-monetary WTP:** 8–12 hours/month saved on manual site visits (valued at S\$200–400 opportunity cost).

At 500 paying Starter subscribers, annual recurring revenue (ARR) = S\$174,000; at 2,000 subscribers, ARR = S\$696,000.

3 Go-to-Market Difficulties

3.1 Trust and Adoption

Customers may distrust a startup’s scores over their own site visits. **Mitigation:** Source transparency (link every score to data.gov.sg records), publish methodology, and partner with NEA CoC training providers for distribution.

3.2 Data Acquisition Costs and Risks

- **NEA/SFA licensing data:** Free, open government data (Open Data Licence). Low acquisition cost.
- **AMR rent benchmarks:** Manually curated from NEA tender PDFs; not available via API. Requires periodic manual updates.
- **MRT proximity:** Free from LTA/data.gov.sg. GeoJSON download available.
- **Risk:** data.gov.sg rate limits (HTTP 429) during bulk ingestion; mitigated with backoff and batch scheduling.

3.3 Legal and Privacy

Licensed establishment data contains business names and addresses but not personal data of individuals. PDPA/GDPR risk is low for B2B aggregated analytics. We do not scrape social media profiles or track individuals. Terms of service for data.gov.sg permit commercial use under the Singapore Open Data Licence.

3.4 Cold-Start Problem

Scores improve with more granular competition data (cuisine type, operating hours). Initial MVP uses postal-prefix aggregation, which is coarse. Chicken-and-egg: operators want proven accuracy before paying; accuracy improves with more paying customers providing feedback.

3.5 Competition and Moats

- **Incumbents:** Manual consultants (high touch, high price); NEA calculator (free, limited scope); Google Maps (reviews, no structured scoring).
- **Moat:** Proprietary scoring model tuned on Singapore hawker economics; curated AMR dataset; integration with NEA tender calendar (future feature).
- **Threat:** Google or Grab could add location scores; differentiation via hawker-specific features and local rent data.

3.6 Unit Economics

Table 3: Unit economics sketch (monthly, at steady state)

Scale	Subscribers	Revenue	Infra Cost
Initial	50	S\$1,450	S\$50 (Vercel + Neon)
10×	500	S\$14,500	S\$200
100×	5,000	S\$145,000	S\$1,500

Gross margin exceeds 85% at scale; primary costs are data curation labor (AMR updates) and customer acquisition, not compute.

4 System Design

4.1 Overview

StallPulse implements a classic **batch analytics pipeline** with a **web delivery layer**:

1. **Ingestion** — Python scripts fetch data from data.gov.sg REST APIs.
2. **Storage** — Raw CSV/JSON in local/cloud object storage; processed Parquet for analytics.
3. **Processing** — Pandas batch job computes viability scores (Spark-compatible design for scale).
4. **Delivery** — Next.js app on Vercel serves dashboard and REST API.

4.2 Data Sources

Table 4: Data sources and ingestion method

Dataset	Source	Method	Rows
Licensed eating establishments	data.gov.sg / NEA	REST API (paginated)	36,687
MRT station names	data.gov.sg / LTA	REST API	184
Hawker centre AMR rents	NEA tender notices	Curated CSV	25

4.3 Storage and Processing

- **Raw layer:** CSV and JSON files (`data/raw/`) — analogous to a distributed file system (HDFS/S3) for immutable raw dumps.
- **Processed layer:** Apache Parquet (`data/processed/`) — columnar format for efficient analytics queries; production would use PostgreSQL (Neon) or BigQuery.
- **Processing:** Python/Pandas batch job (`scripts/process/score_locations.py`). At 100× scale, migrate to Apache Spark on Dataproc or EMR for parallel postal-prefix aggregation across 36K+ records.
- **Message queue (future):** Apache Kafka would trigger re-scoring when data.gov.sg publishes monthly NEA updates, decoupling ingestion from processing.

4.4 Scoring Model

Viability score (0–100) is a weighted composite:

$$S = 0.35 \cdot f_{\text{footfall}} + 0.25 \cdot f_{\text{competition}} + 0.20 \cdot f_{\text{rent}} \\ + 0.10 \cdot f_{\text{hygiene}} + 0.10 \cdot f_{\text{density}}$$

where f_{footfall} is derived from MRT distance, $f_{\text{competition}}$ from licensed establishment density by postal prefix, f_{rent} from AMR relative to market median, f_{hygiene} from average NEA grade in the area, and f_{density} from estimated stall capacity.

4.5 Delivery

- **Web dashboard:** Next.js 16 + React + Tailwind CSS — interactive ranked list with score breakdown.
- **REST API:** GET `/api/locations?area=&min_score=&q=` — JSON feed for Pro subscribers.
- **Future:** PDF location reports, NEA tender alert emails, embedded widget for CoC course portals.

4.6 Architecture Diagram

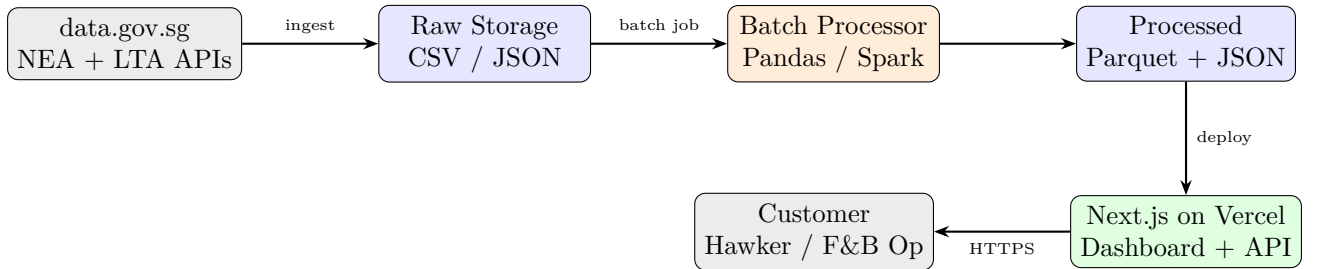


Figure 1: StallPulse end-to-end architecture

4.7 Scalability and Cost at 10× and 100×

10× (500 subscribers, 250 centres):

- Migrate batch processing to Apache Spark (local → 4-node cluster).

- Store processed data in PostgreSQL (Neon serverless) instead of static JSON.
- Add Redis cache for API hot queries.
- Estimated infra: S\$200/month.

100× (5,000 subscribers, 1,000+ centres):

- Stream NEA updates via Kafka; incremental scoring with Spark Structured Streaming.
- Data lake on S3/GCS with Parquet partitioning by planning area.
- CDN for static reports; API rate limiting via API gateway.
- Estimated infra: S\$1,500/month vs. S\$145,000 MRR (99% gross margin).

5 Repository and Deployment

The project repository contains:

- `scripts/ingest/` — data.gov.sg ingestion scripts
- `scripts/process/` — batch scoring pipeline
- `scripts/demand-research/` — demand evidence collection
- `web/` — Next.js delivery application
- `data/` — raw and processed sample data
- `report/report.tex` — this document

Deploy the web app to Vercel by pointing the project root to `web/`. See `VERCEL.md` for step-by-step instructions.

6 Conclusion

StallPulse demonstrates that public government data, when refined through a batch analytics pipeline and delivered via a focused SaaS product, can address a real and monetizable need. The hawker stall entrepreneur segment has clear pain (location selection under uncertainty), validated market size (36K+ establishments), and willingness to pay (S\$29–99/month) anchored against alternatives costing 100× more. Technical risks (rate limits, coarse geocoding) and business risks (trust, cold-start) are manageable with transparent sourcing and iterative model improvement.